

Edunet Foundation

Green Skills Using AI Technologies

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“ Building a sustainable future
through AI, innovation and green skills. ”



2. THINGS I LEARNED FROM THE LINKS

Key insights and learnings from the resources provided



	1. Data Centres (YouTube)	Learned how data centres power our digital world and AI technologies, but consume large amounts of electricity and water for cooling. Using renewable energy, efficient cooling systems and Green AI practices can significantly reduce their environmental impact.	
	2. Ecological Footprint Calculator	Helped me understand how our lifestyle uses Earth's resources. It calculates the land and natural resources required by an individual and shows how we can reduce our ecological footprint and live more sustainably.	
	3. Carbon Footprint Calculator (conservation.org)	Helped estimate carbon emissions from daily activities like transport, home energy use, food and travel. It gave me awareness of my carbon footprint and ways to reduce it by making low-carbon choices.	
	4. Emission Calculator (EcoTransit)	Allowed me to compare CO₂ emissions across different transport modes (road, rail, air, sea). It helps in choosing environmentally friendly logistics and travel options to lower emissions.	
	5. MIT En-ROADS Climate Simulator	Interactive simulator that shows how different policies (renewable energy, carbon pricing, energy efficiency, electrification, etc.) impact global CO₂ emissions and temperature. It helped me understand the importance of collective action for climate goals.	
	6. AI Waste Monitoring Platform (NANDO)	Showed how AI and computer vision can monitor waste in real-time, improve recycling rates, optimize waste collection routes, reduce operational costs and lower CO₂ emissions.	
	7. Jobs SCGJ – Skill Council for Green Jobs (YouTube Channel)	Introduced various green job roles and the importance of green skills in sectors like renewable energy, solar, EVs, environmental management and sustainability to build a green economy.	
	8. CO₂ Calculator (e-AMRIT, NITI Aayog)	Calculates CO₂ emissions of conventional vehicles and compares them with electric vehicles, showing how switching to EVs can significantly reduce emissions and support India's clean mobility mission.	



Slide 3: Activities and Tasks Completed During Internship



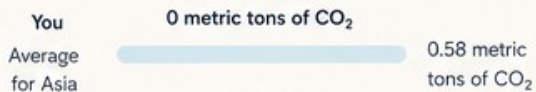
Screenshots of Tools Used – CO₂ Emission Calculator, Emission Calculator & Footprint Calculator



1. Carbon Footprint Calculator (footprint.conservancy.org)

How do you compare?

Here is your annual carbon footprint.

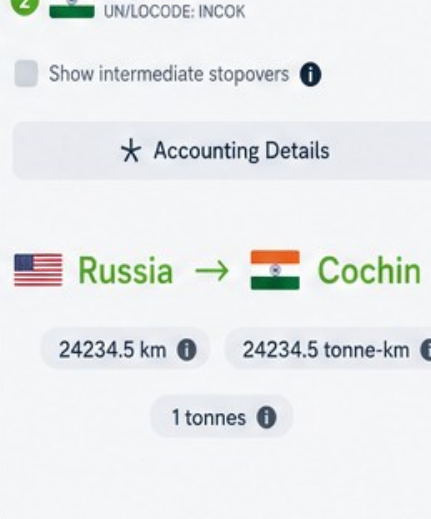


2. Emission Calculator (EcoTransIT World) Route: Russia → Cochin (India)

The interface shows a route from Russia to Cochin, India. It includes fields for origin (US, Russia), destination (IN, Cochin), and a 'Calculate' button. A map below shows the route across the Atlantic and Indian Oceans.



The interface shows a route from Russia to Cochin, India. It includes fields for origin (Russia), destination (Cochin), and a 'Calculate' button. A map below shows the route across the Atlantic and Indian Oceans.



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3. CO₂ Calculator (e-AMRIT) (e-amrit.niti.gov.in)

0.22

CO₂ emissions from conventional vehicle

0.16

CO₂ emissions due to electric vehicle

0.05

Annual CO₂ emissions savings by opting EV

0.79

CO₂ emissions reductions during the lifecycle of the vehicle

Start a new Calculation



Key Takeaway: These tools helped in understanding carbon footprint, transport emissions and the impact of cleaner alternatives like electric vehicles in reducing CO₂ emissions and promoting sustainability.



GREEN SKILLS, SUSTAINABILITY & AI

Building a Sustainable Future with Smart Technologies



WHAT IS GREEN SKILLING?

Green skilling means acquiring knowledge and abilities needed to promote environmental sustainability and support eco-friendly practices.

- ✓ Renewable Energy
- ✓ Waste Management
- ✓ Energy Efficiency
- ✓ Sustainable Agriculture
- ✓ Green Technology & Innovation



WHAT IS SUSTAINABILITY?

Sustainability is meeting the needs of the present without compromising the ability of future generations to meet their own needs.



ENVIRONMENT

Protect natural resources and ecosystems



SOCIETY

Ensure well-being, equity and inclusion



ECONOMY

Promote responsible growth and resource efficiency

BENEFITS OF AI FOR SUSTAINABILITY



ENERGY OPTIMIZATION

AI helps in smart grid management, predictive maintenance and efficient energy distribution.



WASTE MANAGEMENT

AI-powered systems (like smart bins) can identify, sort and manage waste efficiently, improving recycling and reducing landfill.



SMART AGRICULTURE

AI enables precision farming, crop health monitoring, yield prediction and optimal use of water and fertilizers.



CLIMATE & ENVIRONMENT MONITORING

AI analyzes environmental data to predict weather patterns, monitor air quality and detect climate risks early.



RESOURCE EFFICIENCY

AI helps optimize the use of water, materials and other resources, reducing waste and operational costs.

CHALLENGES OF AI FOR SUSTAINABILITY



HIGH ENERGY CONSUMPTION

Training large AI models requires significant electricity, contributing to higher carbon emissions.



DATA CENTRE CARBON FOOTPRINT

Data centres consume large amounts of energy and water for cooling, impacting the environment.



ELECTRONIC WASTE (E-WASTE)

Rapid technological upgrades lead to increased e-waste, which can harm ecosystems if not managed properly.



RESOURCE-INTENSIVE HARDWARE

Manufacturing AI hardware requires rare minerals, water and energy, causing environmental pressure.



Green skills + AI innovation + Sustainability mindset
can create a cleaner, greener and smarter planet for all.





CAPSTONE PROJECT

AI for Environment, Sustainability & Green Skills



SMART ENERGY CONSUMPTION ANALYZER

AI-Powered System for Predicting Energy Usage and Promoting Sustainable Living

PROBLEM STATEMENT

Many households and organizations are unaware of their energy consumption patterns, leading to excessive electricity usage, higher bills, and increased carbon emissions.

OBJECTIVE

- To develop an AI-powered system that analyzes electricity consumption data.
- Predict future energy usage using machine learning.
- Provide actionable recommendations to reduce energy consumption and promote sustainability.

TECHNOLOGIES USED

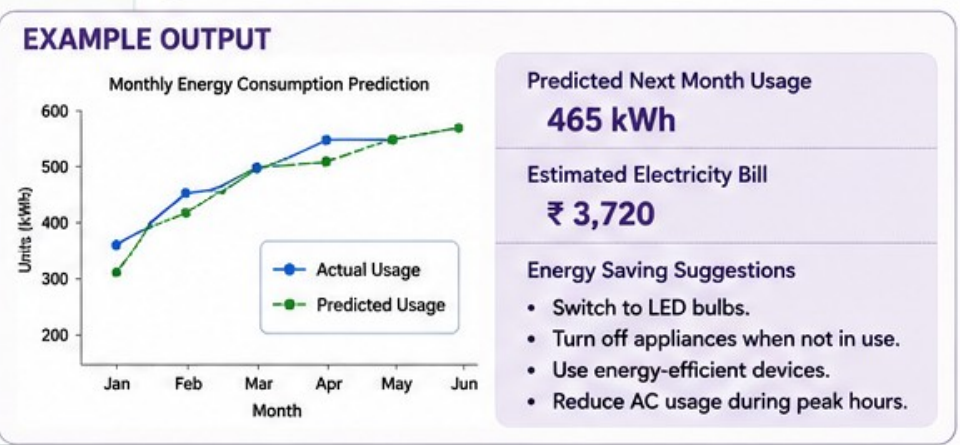
Python Pandas NumPy Scikit-learn Matplotlib Jupyter Notebook

EXPECTED IMPACT

- Reduce electricity wastage.
- Lower electricity bills.
- Increase awareness of sustainable energy use.
- Reduce carbon footprint.
- Support green living and environmental sustainability.

METHODOLOGY

- Collect electricity consumption data.
- Analyze historical usage patterns.
- Train a machine learning model to predict future consumption.
- Generate energy-saving recommendations based on predicted usage.
- Display results using graphs and insightful reports.



SDGS SUPPORTED

7 AFFORDABLE AND CLEAN ENERGY

12 RESPONSIBLE CONSUMPTION AND PRODUCTION

13 CLIMATE ACTION

SDG 17 – PARTNERSHIPS FOR THE GOALS

Collaboration between individuals, organizations, governments and communities is essential to build a sustainable and energy-efficient future.